

Agent Runtime, Tools & Coding Execution

Research objective

- Understand what an actual agentic execution system requires beyond an LLM generating responses.

Research areas

Agent execution

- Agent runtime
- Task decomposition
- Planning
- Execution loop
- Observation
- State
- Intermediate results
- Tool invocation
- Retry
- Error handling
- Self-correction
- Task completion detection
- Termination conditions
- Agent verification

Tool system

- Tool calling
- Tool schemas
- Tool permissions
- Tool authorization
- File tools
- Document tools
- Spreadsheet tools
- Search tools
- Code execution tools
- Verification tools

Coding workflow

Research the complete lifecycle:

Requirement



Plan



Generate/modify code



Execute



Observe result



Run tests



Detect failure



Correct



Re-run



Verify



Deliver

Keywords

Agent Runtime

- agent runtime architecture
- local agent runtime
- offline agent architecture
- air-gapped AI agent
- multi-step agent execution
- agent task decomposition
- agent planning loop
- agent execution loop
- planning reasoning acting loop
- agent state management
- stateful AI agents
- agent workflow orchestration
- agent execution graphs
- agent retry mechanisms
- agent self-correction

- agent intermediate result inspection
- agent task completion detection
- agent termination conditions

Tool Use

- LLM tool calling architecture
- local tool calling
- agent tool orchestration
- controlled tool execution
- tool authorization for AI agents
- agent capability permissions
- agent filesystem tools
- agent document tools
- agent spreadsheet tools
- agent code execution tools
- agent verification tools

Coding Agent

- local coding agent architecture
- software engineering AI agent
- code generation execution verification
- AI generated code testing
- automated code repair agent
- code execution feedback loop
- coding agent sandbox
- repository-aware coding agent
- local software engineering agent
- agentic debugging
- AI code verification
- test-driven coding agent
- code generation test repair loop

Failure research

Investigate:

- Agent planning failures
- Wrong tool selection
- Incorrect tool arguments
- Infinite loops
- Hallucinated tool results
- Failure to verify
- False completion
- Incorrect code generation
- Incorrect test interpretation
- Agent state corruption
- Agent retry failures

Required conclusion

What makes an AI system genuinely agentic rather than merely a chatbot with tool calling? Must be supported by research evidence.